第三章 进程

1. Describe the differences among short-term, medium-term, and long-term scheduling.

① Short-term scheduling: Decides which ready process in memory gets the CPU next. It's the most frequent one.

② Medium-term scheduling: Temporarily suspends or resumes processes. It's less frequent than short-term.

③ Long-term scheduling: Controls which jobs are admitted into the system for processing.

2. Describe the actions taken by a kernel to context-switch between processes.

① Save the state of the current process (register, stack, etc.).

② Update process control structures. Mark the old process as "ready" / "waiting" and so on.

③ Select a new process.

④ Load the state of the new process (PCB, reg, stack, PC, ...)

⑤ Switch memory management context

⑥ Resume execution of the new process.

3. 采用下述程序，确定 A、B、C、D 四行中 pid 和 pid1 的值。（假设父进程和子进程的 pid 分别为 2600 和 2603）

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
   pid_t pid,pid1;

   pid=fork();

   if (pid<0)
   {
      fprintf(stderr,"fork fail");
      return 1;
   }
   else if (pid==0)
   {
        pid1=getpid();
        printf("child:pid=%d",pid);          //A
        printf("child:pid1=%d",pid1);        //B
    }
    else
    {
        pid1=getpid();
        printf("parent:pid=%d",pid);         //C
        printf("parent:pid1=%d",pid1);       //D
        wait(NULL);
    }
   return 0;
}
```

A：0　　B：2603

C：2603　　D：2600